Discrete Applied Mathematics 232 (2017) 1–7

Contents lists available at ScienceDirect

Discrete Applied Mathematics

journal homepage: www.elsevier.com/locate/dam

On the double Roman domination in graphs

Hossein Abdollahzadeh Ahangar [a,*], Mustapha Chellali [b], Seyed Mahmoud Sheikholeslami [c]

[a] Department of Mathematics, Babol Noshirvani University of Technology, Shariati Ave., Babol, Postal Code: 47148-71167, Islamic Republic of Iran
[b] LAMDA-RO Laboratory, Department of Mathematics, University of Blida, B.P. 270, Blida, Algeria
[c] Department of Mathematics, Azarbaijan Shahid Madani University, Tabriz, Islamic Republic of Iran

A R T I C L E I N F O

Article history:
Received 15 January 2017
Received in revised form 24 May 2017
Accepted 13 June 2017
Available online 4 August 2017

Keywords:
Roman domination
Double Roman domination
Roman {2}-domination

A B S T R A C T

A double Roman dominating function (DRDF) on a graph $G = (V, E)$ is a function $f : V(G) \to \{0, 1, 2, 3\}$ having the property that if $f(v) = 0$, then vertex v has at least two neighbors assigned 2 under f or one neighbor w with $f(w) = 3$, and if $f(v) = 1$, then vertex v must have at least one neighbor w with $f(w) \geq 2$. The weight of a DRDF is the value $f(V(G)) = \sum_{u \in V(G)} f(u)$. The double Roman domination number $\gamma_{dR}(G)$ is the minimum weight of a DRDF on G. First we show that the decision problem associated with $\gamma_{dR}(G)$ is NP-complete for bipartite and chordal graphs. Then we present some sharp bounds on the double Roman domination number which partially answer an open question posed by Beeler et al. (2016) in their introductory paper on double Roman domination. Moreover, a characterization of graphs G with small $\gamma_{dR}(G)$ is provided.

1. Introduction

In this paper, G is a simple graph with vertex set $V = V(G)$ and edge set $E = E(G)$. The order $|V|$ of G is denoted by n. For every vertex $v \in V$, the *open neighborhood* $N(v)$ is the set $\{u \in V(G) : uv \in E(G)\}$ and the *closed neighborhood* of v is the set $N[v] = N(v) \cup \{v\}$. The *degree* of a vertex $v \in V$ is $d_G(v) = |N(v)|$. The *minimum* and *maximum degree* of a graph G are denoted by $\delta = \delta(G)$ and $\Delta = \Delta(G)$, respectively. A *leaf* of G is a vertex of degree one, while a *support vertex* of G is a vertex adjacent to a leaf.

We write P_n for the *path* of order n, C_n for the *cycle* of length n and $\overline{K_n}$ for the graph with n vertices and no edges. A *tree* is an acyclic connected graph. A tree T is a *double star* if it contains exactly two vertices that are not leaves. A double star with respectively p and q leaves attached at each support vertex is denoted by $S_{p,q}$. A connected graph G is *unicyclic* if it has a unique cycle. A connected subgraph B of G is a *block*, if B has no cut vertex and every subgraph $B' \subseteq G$ with $B \subsetneq B'$ has at least one cut vertex. A graph G is a *block graph*, if every block is complete. For terminology and notation on graph theory not given here, the reader is referred to [16].

A set $S \subseteq V$ in a graph G is called a *dominating set* if every vertex of G is either in S or adjacent to a vertex of S. The *domination number* $\gamma(G)$ equals the minimum cardinality of a dominating set in G. A subset $S \subseteq V$ is a 2-*dominating set* if every vertex of $V - S$ has at least two neighbors in S. The 2-*domination number* $\gamma_2(G)$ equals the minimum cardinality of a 2-dominating set in G.

* Corresponding author.
E-mail addresses: ha.ahangar@nit.ac.ir (H. Abdollahzadeh Ahangar), m_chellali@yahoo.com (M. Chellali), s.m.sheikholeslami@azaruniv.edu (S.M. Sheikholeslami).

http://dx.doi.org/10.1016/j.dam.2017.06.014

For a graph G and a positive integer k, let $f : V(G) \to \{0, 1, 2, \ldots, k\}$ be a function, and let $(V_0, V_1, V_2, \ldots, V_k)$ be the ordered partition of $V = V(G)$ induced by f, where $V_i = \{v \in V : f(v) = i\}$ for $i \in \{0, 1, \ldots, k\}$. There is a 1–1 correspondence between the functions $f : V \to \{0, 1, 2, \ldots, k\}$ and the ordered partitions $(V_0, V_1, V_2, \ldots, V_k)$ of V, so we will write $f = (V_0, V_1, V_2, \ldots, V_k)$.

A function $f : V(G) \to \{0, 1, 2\}$ is a *Roman dominating function* (RDF) on G if every vertex $u \in V$ for which $f(u) = 0$ is adjacent to at least one vertex v for which $f(v) = 2$. The weight of an RDF is the value $f(V(G)) = \sum_{u\in V(G)} f(u)$. The *Roman domination number* $\gamma_R(G)$ is the minimum weight of an RDF on G. Roman domination was introduced by Cockayne et al. in [7] and was inspired by the work of ReVelle and Rosing [13], and Stewart [14]. Since 2004, a hundred papers have been published on this topic, where several new variations were introduced: weak Roman domination [10], maximal Roman domination [2], mixed Roman domination [3], and very recently Roman {2}-domination [5] and double Roman domination [4], further studied in [1].

A *Roman* {2}*-dominating function* is a function $f : V \to \{0, 1, 2\}$ with the property that for every vertex $v \in V$ with $f(v) = 0$, $f(N(v)) \geq 2$, that is, there is a vertex $u \in N(v)$, with $f(u) = 2$, or there are two vertices $x, y \in N(v)$ with $f(x) = f(y) = 1$. The weight of a Roman {2}-dominating function is the value $f(V) = \sum_{v\in V} f(v)$, and the minimum weight of a Roman {2}-dominating function is called the *Roman* {2}*-domination number* and denoted by $\gamma_{\{R2\}}(G)$. It is worth mentioning that Roman {2}-domination was called in [12] the Italian domination.

As defined in [4], a function $f : V \to \{0, 1, 2, 3\}$ is a *double Roman dominating function* (DRDF) on a graph G if the following conditions hold.

(i) If $f(v) = 0$, then v must have one neighbor in V_3 or at least two neighbors in V_2.

(ii) If $f(v) = 1$, then v must have at least one neighbor in $V_2 \cup V_3$.

The *double Roman domination number* $\gamma_{dR}(G)$ equals the minimum weight of a double Roman dominating function on G, and a double Roman dominating function of G with weight $\gamma_{dR}(G)$ is called a γ_{dR}-function of G.

In this paper, we first show that the decision problem associated with $\gamma_{dR}(G)$ is NP-complete for bipartite and chordal graphs. Then we present some sharp bounds on the double Roman domination number which partially answer an open question posed by Beeler et al. in [4]. Moreover, we give the exact values of the double Roman domination number for paths and cycles, and we provide a characterization of graphs G with small $\gamma_{dR}(G)$.

2. Preliminary results

We make use of the following results in this paper.

Theorem A (*Beeler et al. [4]*)**.** *For any double Roman dominating function f', there exists a double Roman dominating function f of no greater weight than f' for which no vertex is assigned the value* 1.

As defined in [11], for positive integers r and s, let $F_{r,s}$ be the tree obtained from a double star $S_{r,s}$ by subdividing every edge exactly once. Let $\mathcal{F}$ be the family of all such trees $F_{r,s}$; that is, $\mathcal{F} = \{F_{r,s} \mid r, s \geq 1\}$. Also, let $\mathcal{T}$ be the family of trees $T_{k,j}$ of order $k \geq 2$, where $k \geq 2j + 1$ and $j \geq 0$, obtained from a star by subdividing j edges exactly once. We call the pivot vertex of a $T_{k,j}$ the vertex that was the center of the star we started with, prior to subdividing edges.

Theorem B (*Klostermeyer and MacGillivray [12]*)**.** *If T is a non-trivial tree, then $\gamma_{\{R2\}}(T) \geq \gamma(T) + 1$.*

Theorem C (*Henning and Klostermeyer [11]*)**.** *Let T be a non-trivial tree. Then $\gamma_{\{R2\}}(T) = \gamma(T) + 1$ if and only if $T \in \mathcal{F} \cup \mathcal{T}$.*

For the rest of this section, we give the exact values of the double Roman domination number for paths and cycles. Then we characterize all graphs with small double Roman domination number. Recall that the join $G \vee H$ of two graphs G and H is a graph formed from disjoint copies of G and H by connecting each vertex of G to each vertex of H.

Proposition 1. *for $n \geq 1$, $\gamma_{dR}(P_n) = \begin{cases} n & \text{if} \quad n \equiv 0 \pmod 3 \\ n+1 & \text{if} \quad n \equiv 1 \text{ or } 2 \pmod 3. \end{cases}$*

Proof. Let $P := v_1v_2 \ldots v_n$ be a path of order n. Define $h : V(P_n) \to \{0, 1, 2, 3\}$ by $h(v_{3i-1}) = 3$ for $1 \leq i \leq \frac{n}{3}$ and $h(x) = 0$ otherwise if $n \equiv 0 \pmod 3$, by $h(v_n) = 2$, $h(v_{3i-1}) = 3$ for $1 \leq i \leq \frac{n-1}{3}$ and $h(x) = 0$ otherwise when $n \equiv 1 \pmod 3$, and by $h(v_n) = 3$, $h(v_{3i-1}) = 3$ for $1 \leq i \leq \frac{n-2}{3}$ and $h(x) = 0$ otherwise if $n \equiv 2 \pmod 3$. It is easy to verify that h is a DRDF of P_n of weight n if $n \equiv 0 \pmod 3$ and of weight $n + 1$ if $n \not\equiv 0 \pmod 3$. Thus

$$\gamma_{dR}(P_n) \leq \begin{cases} n & \text{if} \quad n \equiv 0 \pmod 3 \\ n+1 & \text{if} \quad n \equiv 1 \text{ or } 2 \pmod 3. \end{cases}$$

Now we prove the inverse inequality. The proof is by induction on n. The statement holds for all paths of order $n \leq 4$. For the inductive hypothesis, let $n \geq 5$ and suppose that for every path of order less than n the result is true. Assume $f = (V_0, V_1, V_2, V_3)$ is a γ_{dR}-function of P_n so that $V_1 = \emptyset$. First let $f(v_n) = 0$. Then we have $f(v_{n-1}) = 3$. Define

$g : V(P_{n-3}) \to \{0, 1, 2, 3\}$ by $g(v_{n-3}) = \min\{3, f(v_{n-2}) + f(v_{n-3})\}$ and $g(v_i) = f(v_i)$ for $i = 1, \ldots, n-4$. Clearly, g is a DRDF of P_{n-3} of weight $\omega(f) - 3$. It follows from the inductive hypothesis that

$$\gamma_{dR}(P_n) = \omega(f) = \omega(g) + 3 \geq \gamma_{dR}(P_{n-3}) + 3 \geq \begin{cases} n & \text{if} \quad n \equiv 0 \pmod 3 \\ n+1 & \text{if} \quad n \equiv 1 \text{ or } 2 \pmod 3). \end{cases}$$

If $f(v_n)+f(v_{n-1}) \geq 3$, then the function $g : V(P_n) \to \{0, 1, 2, 3\}$ defined by $g(v_n) = 0$, $g(v_{n-1}) = 3$ and $g(x) = f(x)$ otherwise, is clearly a γ_{dR}-function of P_n. As above we obtain the inverse inequality. Let $f(v_n)+f(v_{n-1}) \leq 2$. Then we must have $f(v_n) = 2$ and $f(v_{n-1}) = 0$ and this implies that $f(v_{n-2}) \geq 2$. If $f(v_{n-2})+f(v_{n-3}) \geq 3$, then the function $g : V(P_n) \to \{0, 1, 2, 3\}$ defined by $g(v_n) = g(v_{n-2}) = 0$, $g(v_{n-1}) = 3$, $g(v_{n-3}) = \max\{2, f(v_{n-3})\}$ and $g(x) = f(x)$ otherwise, is clearly a γ_{dR}-function of P_n and we can obtain the inverse inequality as above. Let $f(v_{n-2}) + f(v_{n-3}) \leq 2$. Then we have $f(v_{n-2}) = 2$ and $f(v_{n-3}) = 0$. To double Roman dominate v_{n-3}, we must have $f(v_{n-4}) \geq 2$. Define $g : V(P_n) \to \{0, 1, 2, 3\}$ by $g(v_{n-4}) = g(v_{n-1}) = 3$, $g(v_n) = g(v_{n-2}) = g(v_{n-3}) = 0$ and $g(x) = f(x)$ otherwise. It is easy to see that g is a γ_{dR}-function of P_n and as above we can obtain the inverse inequality. Thus

$$\gamma_{dR}(P_n) \geq \begin{cases} n & \text{if} \quad n \equiv 0 \pmod 3 \\ n+1 & \text{if} \quad n \equiv 1 \text{ or } 2 \pmod 3), \end{cases}$$

and the proof is complete. □

Proposition 2. *For* $n \geq 3$, $\gamma_{dR}(C_n) = \begin{cases} n & \text{if} \quad n \equiv 0, 2, 3, 4 \pmod 6 \\ n+1 & \text{if} \quad n \equiv 1, 5 \pmod 6). \end{cases}$

Proof. Let $C_n = (v_1 v_2 \ldots v_n)$ be a cycle on n vertices. Define $h : V(C_n) \to \{0, 1, 2, 3\}$ by $h(v_i) = 2$ for all i odd and $h(x) = 0$ otherwise if $n \equiv 0 \pmod 2$; by $h(v_{3i-1}) = 3$ for $1 \leq i \leq \frac{n}{3}$ and $h(x) = 0$ otherwise if $n \equiv 3 \pmod 6$; and by $h(v_n) = 2$, $h(v_i) = 2$ for all i odd and $h(x) = 0$ otherwise when $n \equiv 1, 5 \pmod 6$. It is easy to verify that h is a DRDF of C_n of weight n if $n \equiv 0, 2, 3, 4 \pmod 6$ and of weight $n+1$ if $n \equiv 1, 5 \pmod 6$. Hence

$$\gamma_{dR}(C_n) \leq \begin{cases} n & \text{if} \quad n \equiv 0, 2, 3, 4 \pmod 6 \\ n+1 & \text{if} \quad n \equiv 1, 5 \pmod 6). \end{cases}$$

To prove the inverse inequality, let f be a γ_{dR}-function of C_n such that $V_1 = \emptyset$. If there are two consecutive vertices v_i and v_{i+1} for some i such that $f(v_i) = f(v_{i+1}) = 0$ or $f(v_i) > 0, f(v_{i+1}) > 0$, then the function f, restricted to $C_n - v_i v_{i+1} = P_n$ is a DRDF and we deduce from Proposition 1 that $\gamma_{dR}(C_n) \geq \gamma_{dR}(P_n)$ which leads to the desired equality. Hence we may assume that vertices of C_n are assigned alternately the values 0 and k, with $k \in \{2, 3\}$. Clearly, in that case n is even and $\gamma_{dR}(C_n) \geq n$, and the proof is complete. □

Proposition 3. *Let G be a connected graph of order $n \geq 3$. Then*

1. $\gamma_{dR}(G) = 3$ *if and only if* $\Delta(G) = n - 1$.
2. $\gamma_{dR}(G) = 4$ *if and only if* $G = \overline{K_2} \vee H$, *where H is a graph with* $\Delta(H) \leq |V(H)| - 2$.
3. $\gamma_{dR}(G) = 5$ *if and only if* $\Delta(G) = n - 2$ *and* $G \neq \overline{K_2} \vee H$ *for any graph H of order* $n - 2$.

Proof. Let $f = (V_0, V_1, V_2, V_3)$ be a γ_{dR}-function of G such that $V_1 = \emptyset$ (by Theorem A). By definition of a DRDF of G, we have $\gamma_{dR}(G) = 2|V_2| + 3|V_3| \geq 3$.

1. If $\Delta(G) = n - 1$, then clearly $\gamma_{dR}(G) = 3$. Conversely, let $\gamma_{dR}(G) = 3$. It follows from $2|V_2| + 3|V_3| = 3$ that $|V_3| = 1$ and $|V_2| = 0$. If $V_3 = \{x\}$, then x must be adjacent to all vertices of $V(G) - \{x\}$ and hence $\Delta(G) = n - 1$.

2. If $G = \overline{K_2} \vee H$, then $\gamma_{dR}(G) \geq 4$ by the first part of this proposition. Now, by assigning 2 to the vertices of $\overline{K_2}$ and 0 to the remaining vertices, we obtain a DRDF of G with weight 4 and hence $\gamma_{dR}(G) = 4$.

Conversely, let $\gamma_{dR}(G) = 4$. We conclude from $2|V_2| + 3|V_3| = 4$ that $|V_2| = 2$ and $|V_3| = 0$. Let $V_2 = \{x, y\}$. By definition, x, y are adjacent to each vertex in $V(G) - \{x, y\}$ and by the first part we have $xy \notin E(G)$. This implies that $G = \overline{K_2} \vee H$ for some graph H of order $n - 2 \geq 2$. Now by item (1), it is clear that $\Delta(H) \leq |V(H)| - 2$.

3. If $\Delta(G) = n - 2$ and $G \neq \overline{K_2} \vee H$ for any graph H of order $n - 2$, then $\gamma_{dR}(G) \geq 5$ by the first and second parts of this proposition. Now, let $x \in V(G)$ be a vertex of degree $n-2$, y be the vertex not adjacent to x, and define $g : V(G) \to \{0, 1, 2, 3\}$ by $g(x) = 3$, $g(y) = 2$ and $g(u) = 0$ for $u \in V(G) - \{x, y\}$. Clearly, g is a DRDF of G with weight 5 and hence $\gamma_{dR}(G) = 5$.

Conversely, let $\gamma_{dR}(G) = 5$. We deduce from $2|V_2| + 3|V_3| = 5$ that $|V_2| = |V_3| = 1$. Let $V_2 = \{y\}$ and $V_3 = \{x\}$. By definition x is adjacent to all vertices in $V(G) - \{y\}$ and so $\Delta(G) = n - 2$. Now the result follows from the second part. □

3. Complexity results

Our aim in this section is to study the complexity of the following decision problem, to which we shall refer as DOUBLE ROM-DOM:

DOUBLE ROM-DOM
Instance: Graph $G = (V, E)$, positive integer $k \leq |V|$.

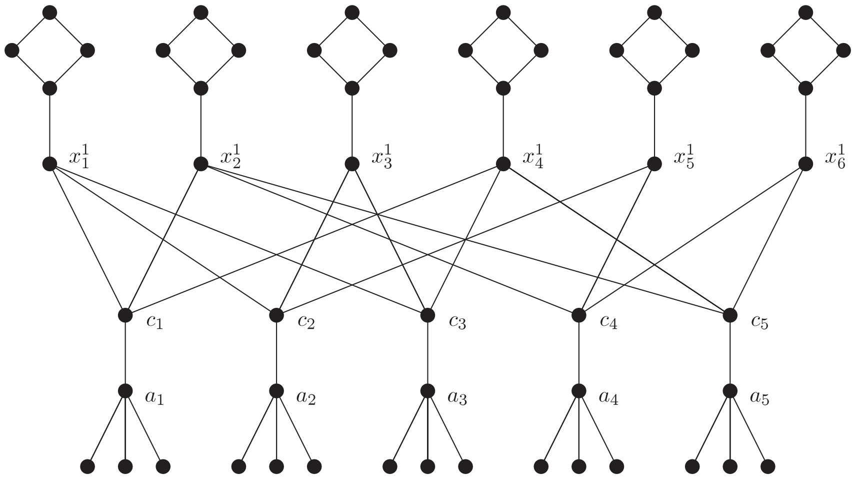

Fig. 1. NP-Completeness for bipartite graphs.

Question: Does G have a double Roman function of weight at most k?

We show that this problem is NP-complete by reducing the well-known NP-complete problem, Exact-3-Cover (X3C), to DOUBLE ROM-DOM.

EXACT 3-COVER (X3C)
Instance: A finite set X with $|X| = 3q$ and a collection C of 3-element subsets of X.
Question: Is there a subcollection C' of C such that every element of X appears in exactly one element of C'?

Theorem 4. *Problem DOUBLE ROM-DOM is NP-Complete for bipartite graphs.*

Proof. DOUBLE ROM-DOM is a member of $\mathcal{NP}$, since we can check in polynomial time that a function $f : V \to \{0, 1, 2, 3\}$ has weight at most k and is a double Roman dominating function. Now let us show how to transform any instance of X3C into an instance G of DOUBLE ROM-DOM so that one of them has a solution if and only if the other one has a solution. Let $X = \{x_1, x_2, \ldots, x_{3q}\}$ and $C = \{C_1, C_2, \ldots, C_t\}$ be an arbitrary instance of X3C.

For each $x_i \in X$, we build a graph H_i obtained from a path $P_5 : x_i^1\text{-}x_i^2\text{-}x_i^3\text{-}x_i^4\text{-}x_i^5$ by adding the edge $x_i^2x_i^5$. For each $C_j \in C$, we build a star $K_{1,4}$ centered at a_j for which one leaf is labeled c_j. Let $Y = \{c_1, c_2, \ldots, c_t\}$. Now to obtain a graph G, we add edges $c_jx_i^1$ if $x_i^1 \in C_j$. Clearly G is bipartite (for example, see Fig. 1). Set $k = 3t + 14q$. Let H be the subgraph of G induced by the union of the $V(H_i)$. Observe that for every DRDF f on G, $f(V(H_i)) \geq 4$ to double Roman dominate all vertices on each cycle $C_4 = (x_i^2, x_i^3, x_i^4, x_i^5)$. Moreover, since H_i has a double Roman domination number equal to 5, we can assume that $f(V(H_i)) \leq 5$. More precisely, if $f(V(H_i)) = 5$, then we may assume, without loss of generality, that $f(x_i^2) = 3, f(x_i^4) = 2$ and $f(x_i^1) = f(x_i^3) = f(x_i^5) = 0$. Also, if $f(V(H_i)) = 4$, then clearly at least one vertex of H_i (including x_i^1) is not double Roman dominated. In this case, we may assume that vertices of H_i are assigned as follows so that only x_i^1 is not double Roman dominated: $f(x_i^2) = f(x_i^4) = 2$ and $f(x_i^1) = f(x_i^3) = f(x_i^5) = 0$.

Suppose that the instance X, C of X3C has a solution C'. We construct a double Roman dominating function f on G of weight k. For every C_j, assign the value 2 to c_j if $C_j \in C'$ and 0 if $C_j \notin C'$. Assign 3 to every a_j and 0 to each leaf neighbor of a_j. Finally, for every i, assign 2 to x_i^2, x_i^4, and 0 to the remaining vertices of H_i. Thereby since C' exists, its cardinality is precisely q, the number of c_j's with weight 2 is q, having disjoint neighborhoods in $\{x_1^1, x_2^1, \ldots, x_{3q}^1\}$, where every x_i^1 has two neighbors assigned 2. Hence, it is straightforward to see that f is a double Roman dominating function with weight $f(V) = 3t + 2q + 12q = k$.

Conversely, suppose that G has a double Roman dominating function with weight at most k. Among all such functions let $g = (V_0, V_1, V_2, V_3)$ be one with $V_1 = \emptyset$ (by Theorem A). Clearly, each star needs a weight of at least 3, and so we may assume, without loss of generality, that $g(a_j) = 3$ and all its leaves are assigned 0. Since $a_jc_j \in E(G)$, it follows that each vertex c_j may be assigned the value 0. Moreover, as mentioned above, for each $i, g(V(H_i)) \in \{4, 5\}$. We may assume that the vertices of H_i are assigned the values given in the second paragraph depending on whether $g(V(H_i)) = 4$ or $g(V(H_i)) = 5$. Let p be the number of H_i's having weight 5. Hence $g(V(H)) = 5p + 4(3q - p) = 12q + p$. Now, if $g(c_j) > 0$ for some j, then c_j serves to double Roman dominate some vertex x_r^1, and in that case $g(c_j) = 2$ (since $g(x_i^2) = 2$). Let y be the number of c_j's assigned 2. Then $3t + 2y + 12q + p \leq k$, implies that $2y + p \leq 2q$. On the other hand, since each c_j has exactly three neighbors in $\{x_1^1, x_2^1, \ldots, x_{3q}^1\}$, we must have $3y \geq 3q - p$. Combining these two inequalities, we arrive at $y = q$ and $p = 0$. Consequently, $C' = \{C_j : g(c_j) = 2\}$ is an exact cover for C. $\square$

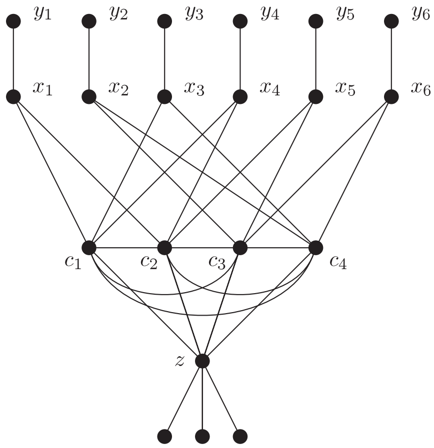

Fig. 2. NP-Completeness for chordal graphs.

Theorem 5. *Problem DOUBLE ROM-DOM is NP-Complete for chordal graphs.*

Proof. Clearly, DOUBLE ROM-DOM is a member of $\mathcal{NP}$. Now let us show how to transform any instance of X3C into an instance G of DOUBLE ROM-DOM so that one of them has a solution if and only if the other one has a solution. Let $X = \{x_1, x_2, \ldots, x_{3q}\}$ and $C = \{C_1, C_2, \ldots, C_t\}$ be an arbitrary instance of X3C.

For each $x_i \in X$, we create a path P_2: x_i-y_i. Let $A = \{x_1, x_2, \ldots, x_{3q}\}$ and $Y = \{y_1, y_2, \ldots, y_{3q}\}$. For each $C_j \in C$, we create a single vertex c_j and adding all edges between c_j's. Next add a star $K_{1,3}$ centered at z by joining z to every c_j. Now to obtain a graph G, we add edges c_jx_i if $x_i \in C_j$. Let $B = \{c_1, c_2, \ldots, c_t\}$. Clearly, G is a chordal graph (for example, see Fig. 2). Set $k = 8q + 3$.

Suppose that the instance X, C of X3C has a solution C'. We construct a DRDF f on G of weight k. For every C_j, assign the value 2 to c_j if $C_j \in C'$ and 0 if $C_j \notin C'$. Assign 3 to z, 2 to every y_i and 0 to the remaining vertices. Since C' exists, its cardinality is precisely q, the number of c_j's with weight 2 is q, having disjoint neighborhoods in $\{x_1, x_2, \ldots, x_{3q}\}$, where every x_i has exactly two neighbors assigned 2. Hence, it is straightforward to see that f is a DRDF on G with weight $f(V) = 3+6q+2q = k$.

Conversely, suppose that G has a DRDF with weight at most k. Among all such functions let $g = (V_0, V_1, V_2, V_3)$ be one with $V_1 = \emptyset$ (by Theorem A) chosen so that:

(C1) $|Y \cap V_3|$ is minimized.

(C2) Subject to Condition (C1): $|B \cap V_3|$ is minimized.

We make few remarks.

(i) Without loss of generality, $g(z) = 3$. Hence every neighbor of z can be defended by z, including vertices of B.

(ii) No vertex y_i is assigned a 3. Indeed, if $g(y_i) = 3$ for some i, then reassigning x_i a 3 and y_i a 0 provides a DRDF of g with weight at most k but with less vertices of Y assigned a 3, a contradiction.

(iii) No vertex x_i is assigned a 2.

(iv) No vertex c_j is assigned a 3. Indeed, if $g(c_j) = 3$ for some j, then c_j serves to defend its neighbors in A, say x_r, x_s, x_u that are assigned a 0. Since y_r, y_s and y_u are assigned each a 2, reassigning c_i a 2 instead of 3 provides a DRDF on G with weight at most k but with less vertices of $B \cap V_3$, a contradiction.

(v) $g(\{x_i, y_i\}) \geq 2$ for each i.

According to items (i)–(v), let $|A \cap V_0| = a$ and $|B \cap V_2| = b$. Clearly $|A \cap V_3| = 3q - a$. Then $g(V(G)) = 3 + 2a + 2b + 3(3q - a) \leq k = 3 + 8q$ implies that $a - 2b \geq q$. On the other hand, since each c_j has exactly three neighbors in A, we must have $3b \geq a$ to double Roman dominate vertices of $A \cap V_0$. Combining these two inequalities with the fact that $a \leq 3q$ we obtain $a = 3q$ and $b = q$. Consequently, $C' = \{C_j : g(c_j) = 2\}$ is an exact cover for C. $\square$

4. Bounds

It was proved in [4] that for every graph G, the double Roman domination number is bounded below by twice the domination number and above by thrice the domination number.

Theorem 6 (*Beeler et al. [4]*). *For any graph G, $2\gamma(G) \leq \gamma_{dR}(G) \leq 3\gamma(G)$.*

In what follows we begin by improving the previous lower bound.

Proposition 7. *For every connected graph G,* $\gamma_{dR}(G) \geq \gamma_{\{R2\}}(G) + \gamma(G)$.

Proof. Clearly the result is valid for graphs of order $n \in \{1, 2\}$. Hence let $n \geq 3$. According to Theorem A, let $f = (V_0, V_1, V_2, V_3)$ be a γ_{dR}-function of G with $V_1 = \emptyset$. Note that $V_2 \cup V_3$ dominates V_0 and so $\gamma(G) \leq |V_2| + |V_3|$. Let V_0' be the vertices of V_0 having at least one neighbor in V_3 and let $V_0'' = V_0 - V_0'$. Now define the function g on G as follows: $g(x) = f(x) - 1$ for all $x \in V_2 \cup V_3$ and $g(x) = f(x)$ for all $x \in V_0$. Clearly, g is a Roman $\{2\}$ -dominating function on G, and so $\gamma_{\{R2\}}(G) \leq |V_2| + 2\,|V_3|$. Therefore, $\gamma_{dR}(G) = 2\,|V_2| + 3\,|V_3| = |V_2| + |V_3| + |V_2| + 2\,|V_3| \geq \gamma(G) + \gamma_{\{R2\}}(G)$. □

Since $\gamma_{\{R2\}}(G) \geq \gamma(G)$ for every graph G and $\gamma_{\{R2\}}(T) \geq \gamma(T) + 1$ for every non-trivial tree T, the following corollaries are immediate from Proposition 7.

Corollary 8 (*Beeler et al. [4]*)**.** *For every connected graph G,* $\gamma_{dR}(G) \geq 2\gamma(G)$.

Corollary 9. *Let T be a nontrivial tree. Then*

$$\gamma_{dR}(T) \geq 2\gamma(T) + 1$$

with equality if and only if $T \in \mathcal{T}$.

Proof. By Theorem B and Proposition 7, we have

$$\gamma_{dR}(T) \geq \gamma_{\{R2\}}(T) + \gamma(T) \geq (\gamma(T) + 1) + \gamma(T) = 2\gamma(T) + 1.$$

If the equality holds, then the above inequality chain leads to $\gamma_{\{R2\}}(T) = \gamma(T) + 1$. Since $\gamma_{dR}(T) = 2\gamma(T) + 2$ if $T \in \mathcal{F}$, we conclude from Theorem C that $T \in \mathcal{T}$.

Conversely, let $T \in \mathcal{T}$. Then the function f that assigns 3 to the pivot vertex, 2 to the leaves at distance 2 from the pivot vertex and 0 to the remaining vertices, is a DRDF of G with weight $2\gamma(T) + 1$ and this implies that $\gamma_{dR}(T) = 2\gamma(T) + 1$. □

It is worth mentioning that Beeler et al. [4] showed that $\gamma_{dR}(G) = 2\gamma(G)$ if and only if $\gamma_2(G) = \gamma(G)$. Moreover, they proved that for every nontrivial tree T, $\gamma_{dR}(T) \geq 2\gamma(T) + 1$. This last result can be easily extended to the class of block graphs (which includes trees) and for unicyclic graphs different from a cycle C_4 by using the following result due to Hansberg and Volkmann [8] and [9].

Theorem 10 (*Hansberg and Volkmann [8,9]*)**.** *If G is a nontrivial block graph or a unicyclic graph different from a cycle* C_4*, then* $\gamma_2(G) \geq \gamma(G) + 1$.

Thereby, we have the following result.

Proposition 11. *If G is a nontrivial block graph or a unicyclic graph different from a cycle* C_4*, then* $\gamma_{dR}(G) \geq 2\gamma(G) + 1$.

Proposition 12. *For any graph G of order n with maximum degree* Δ*,*

$$\gamma_{dR}(G) \geq \frac{2n}{\Delta} + \frac{\Delta - 2}{\Delta}\gamma(G).$$

This bound is sharp for even cycles and paths of order $3k$.

Proof. Let $f = (V_0, V_1, V_2, V_3)$ be a γ_{dR}-function of G. By Theorem A, we may assume that $V_1 = \emptyset$. Let $S = V_0 \cap N(V_3)$ and $T = V_0 - S$. Since each vertex in V_3 dominates at most Δ vertices of S, we have $|S| \leq \Delta|V_3|$. On the other hand, sine each vertex in V_2 dominates at most Δ vertices of T and since each vertex in T has at least two neighbors in T, we have $2|T| \leq |E(V_2, T)| \leq \Delta|V_2|$ yielding $|T| \leq \frac{\Delta}{2}|V_2|$. Hence $|V_0| = |S| + |T| \leq \Delta|V_3| + \frac{\Delta}{2}|V_2|$. Now we have

$$\begin{aligned}
\Delta\gamma_{dR}(G) &= \Delta(2|V_2| + 3|V_3|) \\
&= \Delta(|V_2| + |V_3|) + 2\Delta|V_3| + \Delta|V_2| \\
&\geq \Delta(|V_2| + |V_3|) + 2|V_0| \\
&= (\Delta - 2)(|V_2| + |V_3|) + 2n \\
&\geq (\Delta - 2)\gamma(G) + 2n, \qquad \text{(since } V_2 \cup V_3 \text{ is a dominating set of } G)
\end{aligned}$$

that leads to the desired bound. □

We will now focus on providing two upper bounds on the double Roman domination number for arbitrary graphs G. The first one is in terms of the order of G, the domination and Roman $\{2\}$-dominations numbers, while the second one is in terms of the 2-domination number. These two bounds allow us subsequently to give partial answers to an open question posed by Beeler et al. [4]: For which classes of graphs, or trees, is $\gamma_{dR}(G) \leq n$?

Proposition 13. *For every connected graph G of order* $n \geq 2$*,* $\gamma_{dR}(G) \leq n + \gamma_{\{R2\}}(G) - \gamma(G)$.

Proof. Let $f = (V_0^f, V_1^f, V_2^f)$ be a $\gamma_{\{R2\}}$-function of G such that $|V_2^f|$ is as large as possible. We claim that V_0^f is a dominating set of G. Clearly V_0^f dominates V_2^f. Now, we show that V_0^f dominates V_1^f. Assume, to the contrary, that some vertex $x \in V_1^f$ is not dominated by V_0^f. Since G is connected, let y be any neighbor of x. Define $g : V(G) \to \{0, 1, 2\}$ by $g(x) = 0, g(y) = \max\{2, f(y) + 1\}$ and $g(u) = f(u)$ for $u \in V(G) - \{x, y\}$. Clearly, g is a Roman $\{2\}$-dominating function of G with weight less than $\omega(f)$ or with $|V_2^g| > |V_2^f|$ which is a contradiction. Thus V_0^f dominates V_1^f and so V_0^f is a dominating set of G. Now, consider the function h defined as follows: $h(u) = f(u)$ if $u \in V_0^f$, and $h(u) = f(u) + 1$ if $u \in V_1^f \cup V_2^f$. Clearly, h is a DRDF on G, and hence,

$$\begin{aligned}
\gamma_{dR}(G) &\leq 2\left|V_1^f\right| + 3\left|V_2^f\right| \\
&= \gamma_{\{R2\}}(G) + \left|V_1^f\right| + \left|V_2^f\right| \\
&= \gamma_{\{R2\}}(G) + n - \left|V_0^f\right| \\
&\leq \gamma_{\{R2\}}(G) + n - \gamma(G),
\end{aligned}$$

as desired. □

Corollary 14. *For any connected graph G of order $n \geq 2$ with $\gamma_{\{R2\}}(G) = \gamma(G)$, $\gamma_{dR}(G) \leq n$.*

Proposition 15. *For every graph G, $\gamma_{dR}(G) \leq 2\gamma_2(G)$.*

Proof. Let D be a $\gamma_2(G)$-set. Then assigning a 2 to every vertex of D and a 0 to every vertex not in D provides a double Roman dominating function on G. It follows that $\gamma_{dR}(G) \leq 2\,|D| = 2\gamma_2(G)$. □

Clearly, the bound of Proposition 15 is an improvement on the bound of $3\gamma(G)$ for all graphs with $\gamma_2(G) \leq \frac{3}{2}\gamma(G)$. Moreover, Beeler et al. [4] noticed that $9n/8$ is an upper bound for the double Roman domination number for all graphs G with minimum degree at least three and they were wondering if this bound can be improved. As we shall see, a positive answer will be given as well as for the previous question by using the following result of Chen and Zhou [6] and Proposition 15.

Theorem 16 (*Chen and Zhou [6]*). *If G is a graph of order n and minimum degree $\delta \geq 3$, then $\gamma_2(G) \leq n/2$.*

Corollary 17. *If G is a graph of order n and minimum degree $\delta \geq 3$, then $\gamma_{dR}(G) \leq n$.*

We note that the previous bound is sharp for the complement of the cycle C_6.

Acknowledgments

The authors are grateful to anonymous referees for their remarks and suggestions that helped to improve the manuscript. H. Abdollahzadeh Ahangar acknowledges Babol Noshirvani University of Technology, Iran for a research grant.

References

[1] H. Abdollahzadeh Ahangar, J. Amjadi, M. Atapour, M. Chellali, S.M. Sheikholeslami, Double Roman trees, Ars Combin., to appear.
[2] H. Abdollahzadeh Ahangar, A. Bahremandpour, S.M. Sheikholeslami, N.D. Soner, Z. Tahmasbzadehbaee, L. Volkmann, Maximal Roman domination numbers in graphs, Util. Math. 103 (2017) 245–258.
[3] H. Abdollahzadeh Ahangar, T.W. Haynes, J.C. Valenzuela-Tripodoro, Mixed Roman domination in graphs, Bull. Malays. Math. Sci. Soc. (2015). http://dx.doi.org/10.1007/s40840-015-0141-1.
[4] R.A. Beeler, T.W. Haynes, S.T. Hedetniemi, Double Roman domination, Discrete Appl. Math. 211 (2016) 23–29.
[5] M. Chellali, T.W. Haynes, S.T. Hedetniemi, A. MacRae, Roman {2}-domination, Discrete Appl. Math. 204 (2016) 22–28.
[6] B. Chen, S. Zhou, Upper bounds for f-domination number of graphs, Discrete Math. 185 (1998) 239–243.
[7] E.J. Cockayne, P.A. Dreyer, S.M. Hedetniemi, S.T. Hedetniemi, Roman domination in graphs, Discrete Math. 278 (2004) 11–22.
[8] A. Hansberg, L. Volkmann, Characterization of block graphs with equal 2-domination number and domination number plus one, Discuss. Math. Graph Theory 27 (1) (2007) 93–103.
[9] A. Hansberg, L. Volkmann, Characterization of unicyclic graphs with equal 2-domination number and domination number plus one, Util. Math. 77 (2008) 265–276.
[10] M.A. Henning, S.T. Hedetniemi, Defending the roman empire–a new strategy, Discrete Math. 266 (2003) 239–251.
[11] M.A. Henning, W.F. Klostermeyer, Italian domination in trees, Discrete Appl. Math. 217 (2017) 557–564.
[12] W. Klostermeyer, G. MacGillivray, Roman, Italian, and 2-domination, Manuscript, 2016.
[13] C.S. ReVelle, K.E. Rosing, Defendens imperium romanum: a classical problem in military strategy, Amer. Math. Monthly 107 (7) (2000) 585–594.
[14] I. Stewart, Defend the Roman empire!, Sci. Amer. 281 (6) (1999) 136–139.
[16] D.B. West, Introduction to Graph Theory, second ed., Prentice Hall, USA, 2001.